

An Outline of Receiver Device or Gesture Device Components

Aakanksha*

Research Scholar, Department of Electronics Engineering, Delhi Technical University, Delhi, India

Abstract

In the new era of computer science and language technology, signal acknowledgment is a varied topic with the aim of taking human motions through mathematical algorithms. Signals can create from any physical motion or state then commonly initiate from the face or hand. Present emphases in the field comprise emotion acknowledgment from the face and hand motion response. Various methods have been completed using cameras and computer vision algorithms to understand sign language. Nevertheless, the sympathy and gratitude of pose, gait, proxemics, and humanoid performances is also the matter of signal acknowledgment methods. Signal acknowledgment can be understood as a way for computers to create to realize human body language, so constructing a comfortable bridge among machines and humans than original text operator interfaces before even GUIs (graphical user interfaces), that is still in the limit the popular of input to keyboard and mouse.

Keywords: actuator, decoder, Graphical user interface, microcontroller, motion response, sign language .

***Author for Correspondence** E-mail: aakankshasisodia@gmail.com

INTRODUCTION

Signal Acknowledgment is the capability of a method to classify and answer to the unlike signals of an individual. The situation can be understood as a mode for computers to activate and understand human body language; thus constructing a comfortable bridge among machines and humans than original text user interfaces or even GUIs (graphical user interfaces), which still limit the popular of input to keyboard and mouse [1–5]. This machinery also has the promising nature to modify the way operators' network with computers by removing input strategies such as joysticks, mouse, keyboards and tolerating the creative body to give indications to the computer over gestures such as finger pointing, hand movement [6–12].

RECEIVERS OR ROBOT

This part contains four modules:

1. Receiver
2. Decoder (HT12D)
3. Process (microcontroller 8051)

4. Actuator (motor driver L293D)

RF Receiver Module (RX)

The RF receiver module will receive the data which is transferred by the gesture device. It is also working as similar to the transmitter module as in Figure 1

- Connect the +vcc pin to the 5-volt terminal
- Connect the ground pin to the ground terminal
- The data pin is then connected to the HT12D (pin-14)
- So that we can get the decoded 4-bit data

Decoder (HT12D)

In a very simple way, we can say that an HT12D converts that serial data into parallel which is received by the rf receiver module. The input data is decoded when no error or unmatched codes are found. A valid transmission is indicated by a high signal at VT pin that is pin no 17 as in Figure 2.

- **Pin 18:** It is used to give the +vcc or biasing to the IC HT12D this pin is connected with the +5 volt

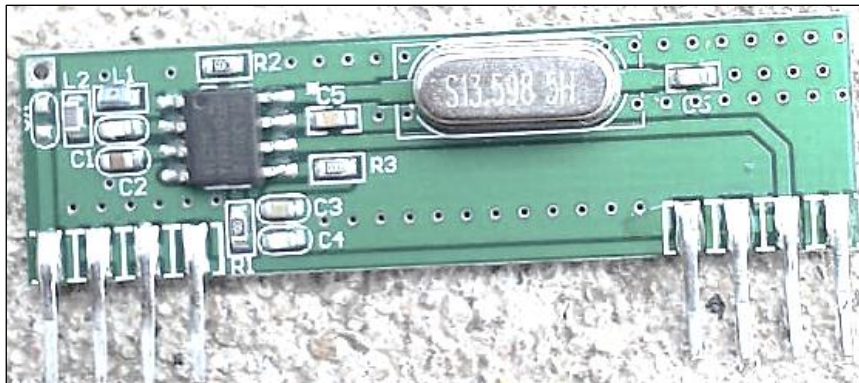


Fig. 1: RF Receiver Module (RX).

- **Pin 17:** It is the valid transmission pin; it will high when the transmission is ok so that we connected this pin to an LED for indication.
- **Pin 16–15:** We connect these two pins directly by a 51 k resistor
- **Pin 14:** This pin is connected with the rf receiver module data pin to receiving the serial data.
- **Pin 10–13:** These pins are data pin which is transferred by the gesture module

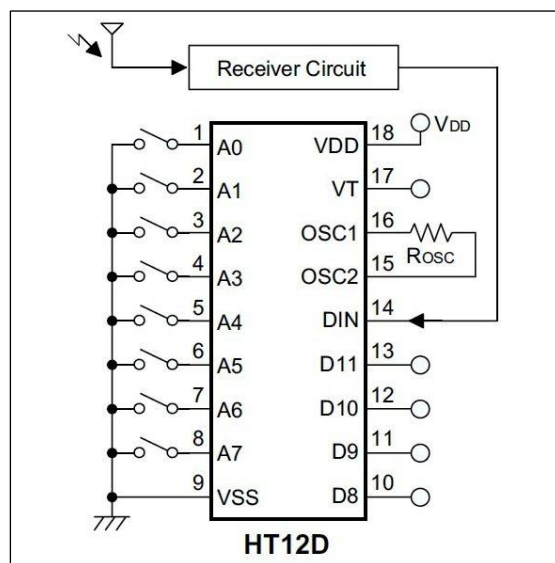


Fig. 2: HT12D Pin Diagram.

Process (Microcontroller P89V51RD2)

The processing is the most important part of the robot. Till now we get the data from the decoder now based on that data; we have to make some decision so here the role of microcontroller is coming up. As in Figure 3, we use an 8051 microcontroller for our circuit to give them a decision capability. Our microcontroller is made up by *npx* the

product name is *P89V51RD2*. The basic circuit to initialize the microcontroller is shown in Figure 3.

We just need a reset circuit and oscillator to run the program; we use the port 1 as an input port and port 2 as an output port. So, the data from the decoder will connect with pin 1, 2, 3, 4 and motor should be connected with pin 21, 22, 23, 24.

For forward the data to the port 2 is 0a or for backward it is 05 then for left its 02 and for right it 08.

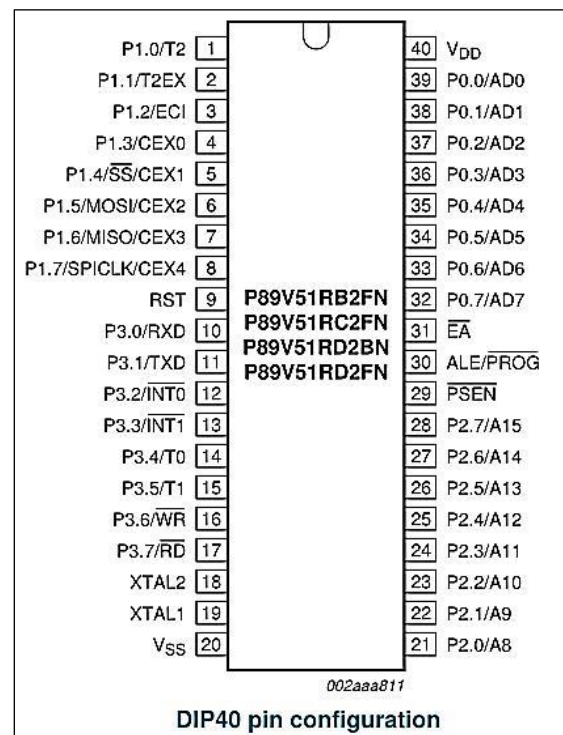


Fig. 3: Microcontroller P89V51RD2 Diagram.

Actuator (L293D)

The actuator are those devices which actually gives the movement or to do a task like motor. In the real world, there are various types of motors available which works on different voltages as Figure 5. So, we need motor driver for running them through the controller. To get interface between motor and microcontroller. We use L293D motor driver IC in our circuit.

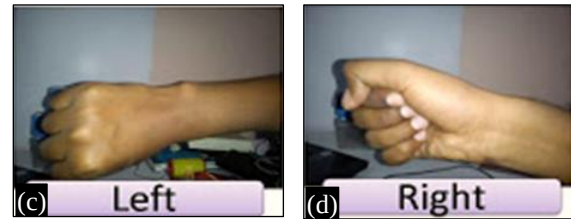
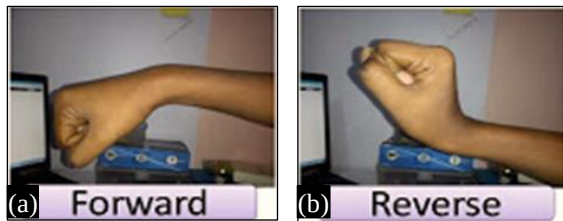


Fig. 4: Hand Movements to Control Accelerometer-based car.

As the circuit in Figure 4, 2 pin male connectors in connected to the pin 8; this will provide the operating voltage for the motor if we want to run our voltage on 12-volt. We have connected a 12-volt power source. The full circuit diagram of the robot (receiver module) the receiver module and receiving module are in Figures 6 and 7.



Fig. 5: Actuator (L293D) Diagram.

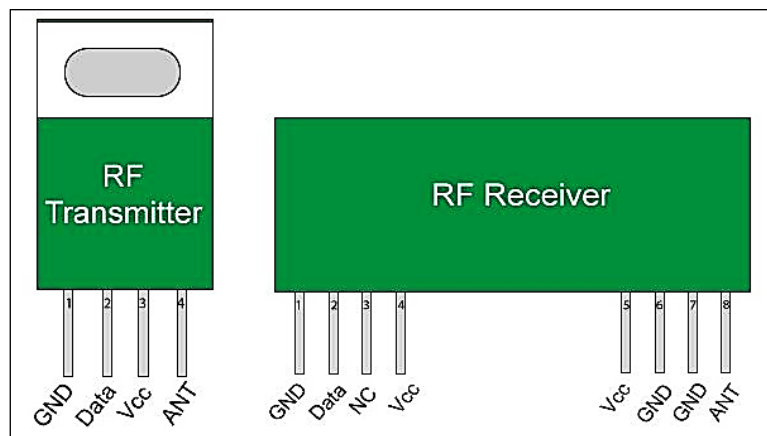


Fig. 6: Receiver Module.

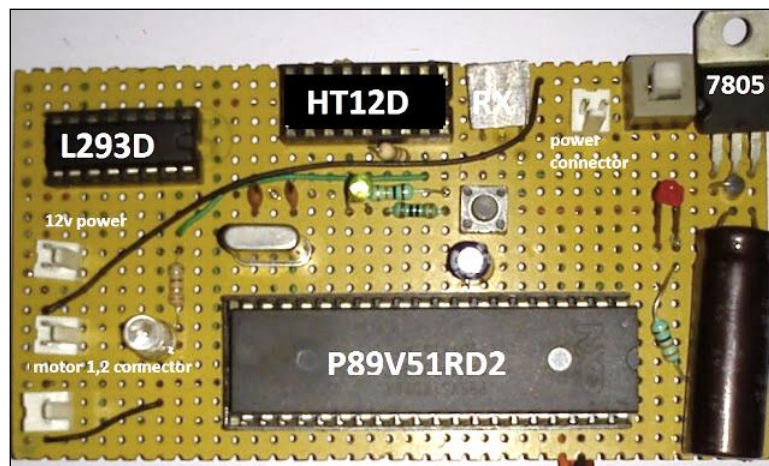


Fig. 7: Actual Circuit of Receiving Module.

PROGRAM

Now, below is the program of the Gesture Controlled robot. The program is in Embedded-C. The microcontroller used is 8051.

/******

Gesture Controlled robot C program using 8051 microcontroller

INPUT DATA to PORT 1 LM324 OUTPUT DATA to PORT 2 for L293D

MSB LSB MSB LSB

1111 0001(0xf1) 0000 1010(0x0a) forward

1111 0010(0xf2) 0000 0010(0x02) right

1111 0100(0xf4) 0000 1000(0x08) left

1111 1000(0xf8) 0000 0101(0x05) backward

*****/

#include<reg51.h> //INCLUDE reg51.h for 8951

void main()

{

P1=0xff; // set port as input port

P2=0x00; // set port as output port

while(1) // infinite loop

{

if(P1==0xf1)

{

P2=0x0a;

}

else if(P1==0xf2)

{

P2=0x02;

}

else if(P1==0xf4)

{

P2=0x08;

}

else if(P1==0xf8)

{

P2=0x05;

}

else

{

P2=0x00;

}

}

APPLICATIONS

- Burglar alarm system
- Smoke and fire alarm system
- Garage door controllers
- Car door controllers
- Car alarm system
- Security system
- Cordless telephones
- Other remote-control systems

CONCLUSION

The variable resistance is obtained across the single terminal at front and one of the two other terminals. The two legs at back offer fixed resistance which is divided by the front leg. So whenever only the back terminals are used, a preset act as a fixed resistor. Presets are specified by their fixed value resistance. For switching applications, transistor is biased so that it remains fully on if there is a signal at its base. In the absence of base signal, it gets completely off.

REFERENCES

1. Starner T, Auxier J, Ashbrook D, Gandy M. The gesture pendant: a self-illuminating, wearable, infrared computer vision system for home automation control and medical monitoring. *Proceedings of the Fourth International Symposium on Wearable Computers, ISWC 2000*, pp. 87–95.
2. Rekimoto J. GestureWrist and gesturepad: unobtrusive wearable interaction devices. *Proceedings of the Fifth International Symposium on Wearable Computers, ISWC, 2001*, pp. 21–31.
3. Sawada H, Hashimoto S. Gesture recognition using an accelerometer sensor and its application to musical performance control. *Electron Commun JPN Part 3*, 2000, pp. 9–17.
4. Hoffman F, Heyer P, Hommel G. Velocity profile based recognition of dynamic gestures with discrete hidden Markov models. *Proceedings of Gesture Workshop '97*, Springer, Berlin Heidelberg, New York, 1997.
5. Tsukada K, Yasumura M. Ubi-finger: gesture input device for mobile use. *Proceedings of APCHI 2002*. 2002;1:388–400.
6. Wilson A, Shafer S. Between u and i: XWand: UI for intelligent spaces. In: *Proceedings of the Conference on Human Factors in Computing Systems, CHI 2003*, April 2003, pp. 545–552.
7. Flanagan J, Mäntylä J, Korpiaho K, Tikanmäki J. Recognizing movements of a handheld device using symbolic representation and coding of sensor signals. *Proceedings of the First International Conference on Mobile and Ubiquitous Multimedia*. 2002, pp. 104–112.
8. Mäntylä V-M, Mäntylä J, Seppänen T, Tuuluri E. Hand gesture recognition of a mobile device user. *Proceedings of the International IEEE Conference on Multimedia and Expo*. 2000, pp. 281–284.
9. Theodoridis S, Koutroumbas K. *Pattern Recognition*. London: Academic Press; 1999.
10. Pirhonen A, Brewster S, Holgiun C. Gestural and audio metaphors as a means of control for mobile devices. *CHI*. April 2002;291–298.
11. Tuuluri E, Ylisaukko-oja A. SoapBox: a platform for ubiquitous computing research and applications. *First International Conference, Pervasive*. 2002;26–28.
12. Rabiner L. Tutorial on hidden Markov models and selected applications in speech recognition. *Proceedings of the IEEE*. 1998;77(2).

Cite this Article

Aakanksha. An Outline of Receiver Device or Gesture Device Components. *Journal of VLSI Design Tools & Technology*. 2019; 9(3): 21–25p.